www.tntextbooks.in

decreases, the average kinetic energy of molecules also decreases, hence the molecular attraction is enhanced. The temperature at which a real gas obeys ideal gas law over an appreciable range of pressure is called Boyle temperature or Boyle point. The Boyle point varies with the nature of the gas. Above the Boyle point, for real gases, $Z > 1$, ie., the real gases show positive deviation. Below the Boyle point, the real gases first show a decrease for Z, reaches a minimum and then increases with the increase in pressure. So, it is clear that at low pressure and at high temperature, the real gases behave as ideal gases.

6.5.2 Compressibility factor for real gases

The compressibility factor Z for real gases can be rewritten

$$Z = \frac{PV_{real}}{nRT} \quad -----(6.17)$$

$$V_{ideal} = \frac{nRT}{P} \quad ------(6.18)$$

Substituting 6.18 in 6.17

$$Z = \frac{V_{real}}{V_{ideal}} \text{ ---------- } (6.19)$$

Where V_{real} is the molar volume of the real gas and V_{ideal} is the molar volume of it when it behaves ideally.

6.5.3 Van der Waals Equation

J. D. Van der Waals made the first mathematical analysis of real gases. His treatment provides us an interpretation of real gas behaviour at the molecular level. He modified the ideal gas equation PV = nRT by introducing two correction factors, namely, pressure correction and volume correction.

Pressure Correction:

The pressure of a gas is directly proportional to the force created by the bombardment of molecules on the walls of the container. The speed of a molecule moving towards the wall of the container is reduced by the attractive forces exerted by its neighbours. Hence, the measured gas pressure is lower than the ideal pressure of the gas. Hence, van der Waals introduced a correction term to this effect.

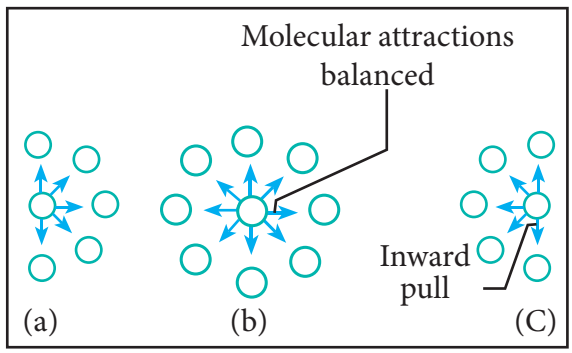


Figure. 6.10 Inter-molecular forces of attraction

Van der Waals found out the forces of attraction experienced by a molecule near the wall are directly proportional to the square of the density of the gas.

$$P' \alpha \rho^2$$

$$\rho = \frac{n}{V}$$